

Solutions to the exercises

Ex 1 pg 213 a) $\Rightarrow$ c) Suppose $f(x)$ is of the form $f(x) = h(x) \cdot k(x)$, we must show $h(x)$ or $k(x)$ is a unit. but $h(x) \cdot k(x) \in (f(x))$

implies say $h(x) \in (f(x))$ so $h(x) = \lambda(x) \cdot f(x)$ so

$f(x) = f(x) \lambda(x) k(x)$ so $f(x) (1 - \lambda(x) k(x)) = 0$ but $k(x)$ is

entire so $\lambda(x) k(x) = 1$ so $k(x)$ is a unit as required.

c) $\Rightarrow$ b) Suppose $(f(x)) \subset I \subset K[x]$. $K[x]$ is a PID so

$I = (g(x))$ so $f(x) = \lambda(x) \cdot g(x) \rightarrow g$ is a unit $\Rightarrow I = K[x] \Rightarrow f(x)$ is maximal ✓
 $\rightarrow \lambda$ is a unit $\Rightarrow I = (f(x))$

b) $\Rightarrow$ a) All maximal ideals are prime

Ex 3 pg 213 $f(x+y) = f(x) + \sum_{i=1}^m f_i(x) y^i$ is clear (see Thm 5.3)

By linearity it is enough to show $f_i(x) = \frac{D^i f(x)}{i!}$ when

$$f(x) = a_n x^n. \quad \frac{D^i(a_n x^n)}{i!} = a_n \frac{n(n-1) \dots (n-i+1)}{i!} x^{n-i}$$

$$a_n (x+y)^n = a_n \left(x^n + \binom{n}{1} x^{n-1} y + \binom{n}{2} x^{n-2} y^2 + \dots + \binom{n}{i} x^{n-i} y^i + \dots \right)$$

Ex 5 a) $(x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$ Both irreducible by Eisenstein's Criterion.
 $(x+1)^6 + (x+1)^3 + 1 = x^6 + 6x^5 + 15x^4 + 21x^3 + 18x^2 + 9x + 3$

b) If a poly of deg 3 is reducible, it has a linear factor and so a root. The only possible roots of $x^3 - 5x^2 + 1$ are ± 1 . So $x^3 - 5x^2 + 1$ is irreducible.

c) Suppose $(x^2 + y^2 - 1)$ is reducible, then write $x^2 + y^2 - 1 = (ax + b)(cx + d)$ with $a, b, c, d \in K[y]$. Then a, c are units so we may assume $x^2 + y^2 - 1 = (x + b)(x + d)$. Then $b + d = 0$ and $bd = -b^2 = y^2 - 1$.

But $-(y^2 - 1)$ is not a square in $K[y]$ as $-(y^2 - 1) = (y - 1)(y + 1)$ (is a square only if $\text{char } K = 2$).

Ex 8 p. 214 $\phi: A[x] \rightarrow A[x] \quad \phi(f(x)) = f(ax+b)$
 $\phi^{-1}(f(x)) = f\left(\frac{x}{a} - \frac{b}{a}\right)$

Ex 9 $\phi: A[x] \rightarrow A[x] \quad \phi(a) = a \quad \forall a \in A, \quad \phi(x) = p(x) \in A[x]$
 $\text{so } \phi(ax^n) = a(p(x))^n. \quad x \in \text{Im } \phi \quad \text{so } x = g(p(x)) \quad \text{some } g(x) \in A[x]$

$\deg g(p(x)) = \deg g \cdot \deg p \quad \text{so } \deg g = \deg p = 1$

$\text{so } p(x) = ax+b, \quad g(x) = cx+d. \quad \text{One easily sees } a^{-1} = c \quad \text{so}$
 $a \text{ is a unit in } A$

Ex 10 $\phi: K(x) \rightarrow K(x) \quad \phi(x) = f(x)/g(x) \quad \phi\left(\frac{p}{q}\right) = \frac{p(f/g)}{q(f/g)}$

ϕ is surjective so there exists $p/q \in K(x)$ such that

$\frac{p(f/g)}{q(f/g)} = x. \quad \text{We may assume } (f,g)=1 \quad (p,q)=1$
 $\text{write } p = a_k x^k + \dots + a_1 x + a_0 \quad q = b_j x^j + \dots + b_1 x + b_0$

Then $\frac{a_k f^k + a_{k-1} f^{k-1} g + \dots + a_0 g^k}{g^k} = x \frac{(b_j f^j + \dots + b_0 g^j)}{g^j}$

$\text{so } g^j (a_k f^k + \dots + a_0 g^k) = x g^k (b_j f^j + \dots + b_0 g^j)$

We claim $j=k$. (If say $k > j$ then $(a_k f^k + \dots + a_0 g^k) = x g^{k-j} (b_j f^j + \dots + b_0 g^j)$
 $\text{so } g \mid a_k f^k$ which is impossible)

$\text{so } a_k f^k + \dots + a_0 g^k = x (b_k f^k + \dots + b_0 g^k)$

$\text{so } (a_k - x b_k) f^k + \dots + (a_0 - x b_0) g^k = 0 \quad \text{so } g \mid (a_k - x b_k) \text{ and } f \mid (a_0 - x b_0)$

$\text{so } g = Ax - B \quad f = Cx - D \quad \text{as required.}$